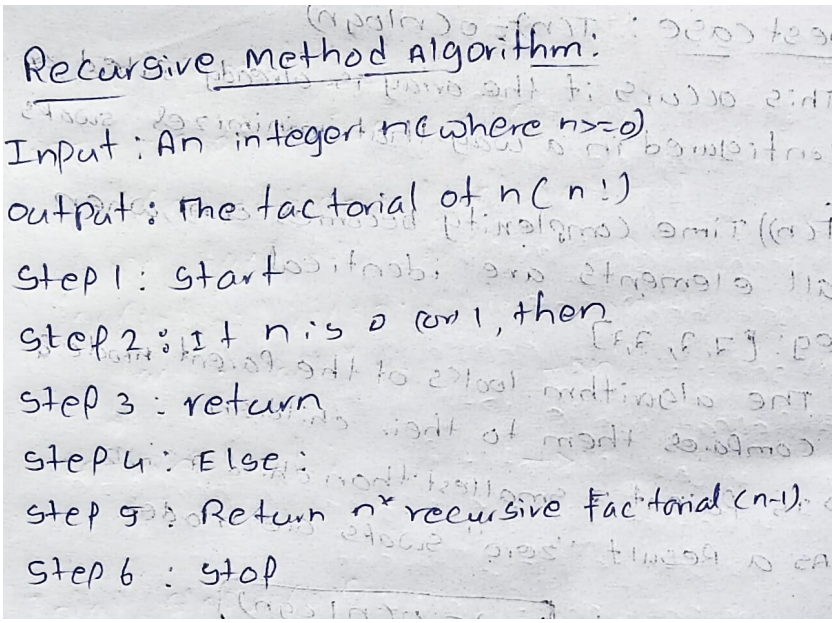
PRACTICAL-4

Aim : To implement and analyze the execution time of the factorial algorithm using both iterative and recursive approaches

1) Recursive Method :



Program / Code :

```
#include <iostream>
using namespace std;

class Factorial
{
public:
   int factr(int n)
   {
      if (n == 0 || n == 1)
      {
         return 1;
      }
      else
      {
         int result = n * factr(n - 1);
         return result;
      }
```

```
    }
};

int main()
{
    int n, result;

      Factorial obj;

    cout << "Enter the value to find its factorial: ";
    cin >> n;

    result = obj.factr(n);

    cout << "Factorial of " << n << " is " << result << endl;

    return 0;
}
```

Output :

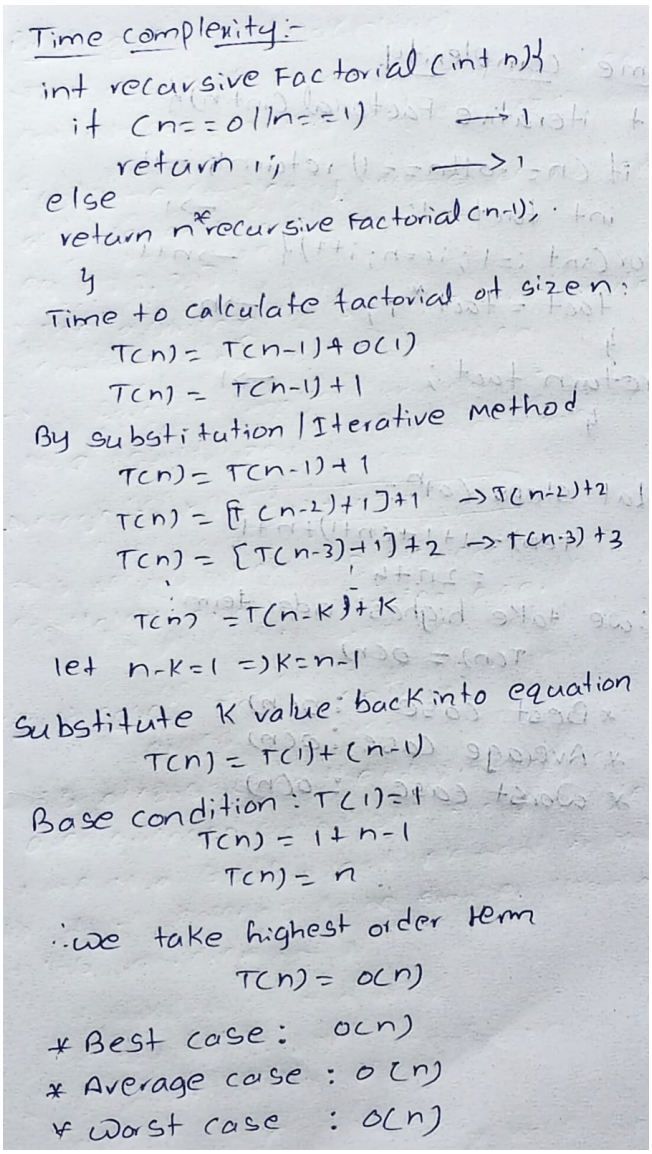
```
Enter the value to find its factorial: 5
Factorial of 5 is 120
```

Time Complexity :

Best Case : O(n)

Average Case : O(n)

Worst Case : O(n)



Time complexity:-

```
int recursive Factorial (int n){
  if (n==0||n==1)        ---> 1
    return 1;            ---> 1
  else
  return n*recursive factorial(n-1);
}
```

Time to calculate factorial of size n:

$T(n) = T(n-1) + O(1)$

$T(n) = T(n-1) + 1$

By substitution / Iterative method

$T(n) = T(n-1) + 1$

$T(n) = [T(n-2)+1]+1 \rightarrow T(n-2)+2$

$T(n) = [T(n-3)+1]+2 \rightarrow T(n-3)+3$

$T(n) = T(n-K)+K$

let $n-K=1 \Rightarrow K=n-1$

Substitute K value back into equation

$T(n) = T(1)+(n-1)$

Base condition: $T(1)=1$

$T(n) = 1+n-1$

$T(n) = n$

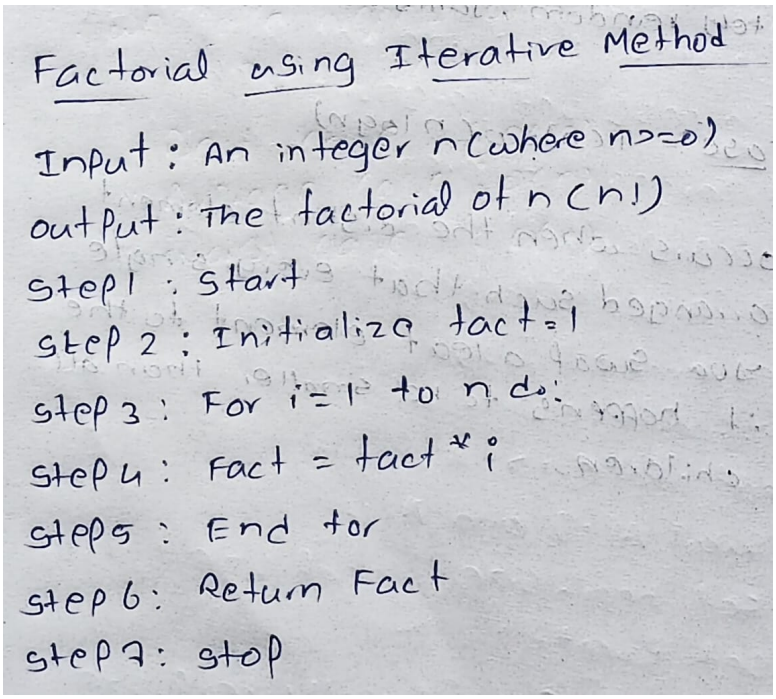
∴ we take highest order term

$T(n) = O(n)$

* Best case: $O(n)$
* Average case: $O(n)$
* Worst case: $O(n)$

2) Iterative Method :

Algorithm :



Program / Code :

```
#include <iostream>
using namespace std;

class Factorial
{
public:
   int fact(int n)
   {
      int result = 1;

      for (int i = 1; i <= n; i++)
      {
         result = result * i;
      }

      return result;
   }
};

int main()
```

```
{
    int n, result;

Factorial obj;

    cout << "Enter the value to find its factorial: ";
    cin >> n;

    result = obj.fact(n);

    cout << "Factorial of " << n << " is " << result << endl;

    return 0;
}
```

Output :

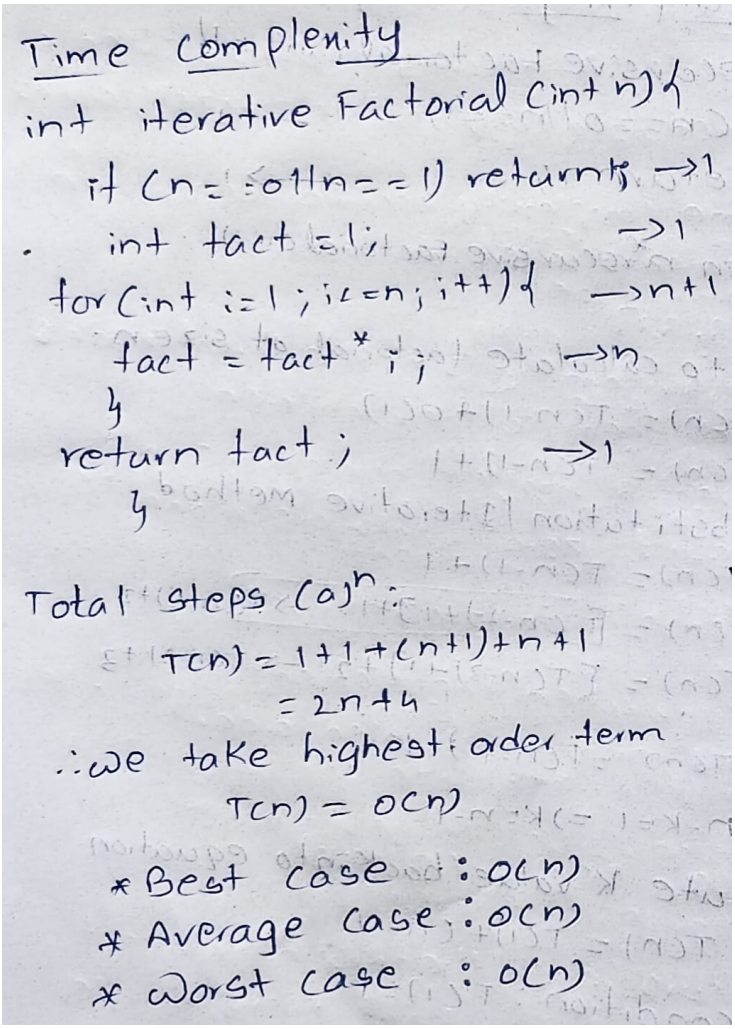
```
Enter the value to find its factorial: 10
Factorial of 10 is 3628800
```

Time complexity :

Best Case : O(n)

Average Case : O(n)

Worst Case : O(n)



Conclusion : The iterative method is faster and more reliable because it uses a direct loop. The recursive method is slower because managing multiple function calls takes more time.